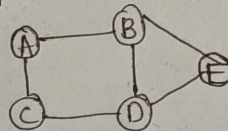


DAA ASSIGNMENT-01

01. a) Write and explain recursive algorithm for binary search.
b) Explain in detail, the solution to four queen problem. using back tracking.
02. What is performance analysis explain space complexity and time complexity with examples.
03. a) Using merge sort the following elements
310, 285, 179, 652, 351, 423, 861, 254, 450, 520.
04. a) Describe the back-tracking technique through the 'n' colouring graph. Explain with an example.
04. a) Sort the record with the following index value in the ascending order using quick sort algorithm.
2, 3, 8, 5, 4, 7, 6, 9, 1.
- b) Heap a set of non-negative integers [10, 7, 5, 18, 12, 20, 15] and a value Sum=35. Explain sum of subset problem. Illustrate the above example.
05. Write an algorithm to determine Hamiltonian cycle in a given graph using backtracking for the following graph.



06. a) What are disjoint set operations. Explain union and find algorithm with example.
- b) Sort the following elements using Heap sort
14, 12, 9, 8, 7, 10, 18.